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IMPERIAL COLLEGE LONDON

DEPARTMENT OF ELECTRICAL AND ELECTRONIC ENGINEERING
EXAMINATIONS

MSc and EEE/EIE PART IV: MEng and ACGI

DIGITAL SIGNAL PROCESSING AND DIGITAL FILTERS

Wednesday, 30 April 2025, 10:00 AM

Time allowed: 3:00 hours

There are FOUR questions on this paper.

Answer Question ONE and any TWO other questions.

Question One is worth 40% of the marks and other questions are worth 30%.

Any special instructions for invigilators and information for candidates are on page 1.

Examiners responsible First Marker(s) : A. Bhandari
Second Marker(s) : T. Stathaki

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Department of Electronic and Electrical Engineering

Examinations 2025

Digital Signal Processing and Digital Filters

MSc and EEE/EIE PART IV

April, 2025 • Exam Duration: Three Hours

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Answer Question 1 and any TWO other questions.

Question 1 is worth 40% of the total marks
while each of the other questions is worth 30%.

If necessary, make reasonable assumptions for your solutions and clearly explain them in your answers.

A formula sheet is attached in the next two pages.

Markers: Ayush Bhandari and Tania Stathaki.

FORMULA SHEET AVAILABLE IN EXAM

The following formulae will be available in the exam:

Where a question requires a numerical answer, it must be given as a fully evaluated decimal number and not as an unevaluated arithmetic expression.

Notation

- All signals and filter coefficients are real-valued unless explicitly noted otherwise.
- Unless otherwise specified, upper and lower case letters are used for sequences and their z-transforms respectively. The signal at a block diagram node V is $v[n]$ and its z-transform is $V(z)$.
- $x[n] = [a, b, c, d, e, f]$ means that $x[0] = a, \ldots x[5] = f$ and that $x[n] = 0$ outside this range.
- $\Re(z)$, $\Im(z)$, z^*, $|z|$ and $\angle z$ denote respectively the real part, imaginary part, complex conjugate, magnitude and argument of a complex number z.
- The expected value of x is denoted $E\{x\}$.
- In block diagrams: solid arrows denote the direction of signal flow; an open triangle denotes a gain element with the gain indicated adjacently; a "+" in a circle denotes an adder/subtractor whose inputs may be labelled "+" or "−" according to their sign; the sample rate, f, of a signal in Hz may be indicated in the form "@f".

Abbreviations

BIBO	Bounded Input, Bounded Output	IIR	Infinite Impulse Response
CTFT	Continuous-Time Fourier Transform	LTI	Linear Time-Invariant
DCT	Discrete Cosine Transform	MDCT	Modified Discrete Cosine Transform
DFT	Discrete Fourier Transform	PSD	Power Spectral Density
DTFT	Discrete-Time Fourier Transform	SNR	Signal-to-Noise Ratio
FIR	Finite Impulse Response		

Standard Sequences

- $\delta[n] = 1$ for $n = 0$ and 0 otherwise.
- $\delta_{\text{condition}}[n] = 1$ whenever "condition" is true and 0 otherwise.
- $u[n] = 1$ for $n \geq 0$ and 0 otherwise.

Geometric Progression

- $\sum_{n=0}^{r} \alpha^n z^{-n} = \frac{1-\alpha^{r+1} z^{-r-1}}{1-\alpha z^{-1}}$ provided that $\alpha z^{-1} \neq 1$.
- $\sum_{n=0}^{\infty} \alpha^n z^{-n} = \frac{1}{1-\alpha z^{-1}}$ provided that $\left|\alpha z^{-1}\right| < 1$.

Forward and Inverse Transforms

z:	$X(z) = \sum_{-\infty}^{\infty} x[n] z^{-n}$	$x[n] = \frac{1}{2\pi j} \oint X(z) z^{n-1} dz$
CTFT:	$X(j\Omega) = \int_{-\infty}^{\infty} x(t) e^{-j\Omega t} dt$	$x(t) = \frac{1}{2\pi} \int_{-\infty}^{\infty} X(j\Omega) e^{j\Omega t} d\Omega$
DTFT:	$X(e^{j\omega}) = \sum_{-\infty}^{\infty} x[n] e^{-j\omega n}$	$x[n] = \frac{1}{2\pi} \int_{-\pi}^{\pi} X(e^{j\omega}) e^{j\omega n} d\omega$
DFT:	$X[k] = \sum_{0}^{N-1} x[n] e^{-j2\pi \frac{kn}{N}}$	$x[n] = \frac{1}{N} \sum_{0}^{N-1} X[k] e^{j2\pi \frac{kn}{N}}$
DCT:	$X[k] = \sum_{n=0}^{N-1} x[n] \cos \frac{2\pi(2n+1)k}{4N}$	$x[n] = \frac{X[0]}{N} + \frac{2}{N} \sum_{n=1}^{N-1} X[k] \cos \frac{2\pi(2n+1)k}{4N}$
MDCT:	$X[k] = \sum_{n=0}^{2N-1} x[n] \cos \frac{2\pi(2n+1+N)(2k+1)}{8N}$	$y[n] = \frac{1}{N} \sum_{0}^{N-1} X[k] \cos \frac{2\pi(2n+1+N)(2k+1)}{8N}$

Convolution

DTFT:	$v[n] = x[n] * y[n] \triangleq \sum_{r=-\infty}^{\infty} x[r] y[n-r]$	$\Leftrightarrow$	$V\left(e^{j\omega}\right) = X\left(e^{j\omega}\right) Y\left(e^{j\omega}\right)$
	$v[n] = x[n] y[n]$	$\Leftrightarrow$	$V\left(e^{j\omega}\right) = \frac{1}{2\pi} X\left(e^{j\omega}\right) \circledast Y\left(e^{j\omega}\right) \triangleq \frac{1}{2\pi} \int_{-\pi}^{\pi} X\left(e^{j\theta}\right) Y\left(e^{j(\omega-\theta)}\right) d\theta$
DFT:	$v[n] = x[n] \circledast_N y[n] \triangleq \sum_{r=0}^{N-1} x[r] y[(n-r)_{\bmod N}]$	$\Leftrightarrow$	$V[k] = X[k] Y[k]$
	$v[n] = x[n] y[n]$	$\Leftrightarrow$	$V[k] = \frac{1}{N} X[k] \circledast_N Y[k] \triangleq \frac{1}{N} \sum_{r=0}^{N-1} X[r] Y[(k-r)_{\bmod N}]$

Group Delay

The group delay of a filter, $H(z)$, is $\tau_H(e^{j\omega}) = -\frac{d\angle H(e^{j\omega})}{d\omega} = \Re\left(\frac{-z}{H(z)} \frac{dH(z)}{dz}\right)\Big|_{z=e^{j\omega}} = \Re\left(\frac{\mathscr{F}(nh[n])}{\mathscr{F}(h[n])}\right)$ where $\mathscr{F}(\,)$ denotes the DTFT.

Order Estimation for FIR Filters

Three increasingly sophisticated formulae for estimating the minimum order of an FIR filter with unity gain passbands:

1. $M \approx \frac{a}{3.5\Delta\omega}$
2. $M \approx \frac{a-8}{2.2\Delta\omega}$
3. $M \approx \frac{a-1.2-20\log_{10} b}{4.6\Delta\omega}$

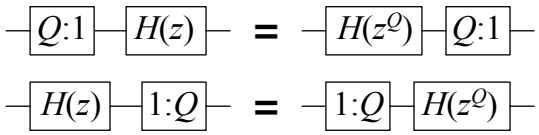
where a =stop band attenuation in dB, b = peak-to-peak passband ripple in dB and $\Delta\omega$ = width of smallest transition band in radians per sample.

z-plane Transformations

A lowpass filter, $H(z)$, with cutoff frequency ω_0 may be transformed into the filter $H(\hat{z})$ as follows:

Target $H(\hat{z})$	Substitute	Parameters
Lowpass $\hat{\omega} < \hat{\omega}_1$	$z^{-1} = \frac{\hat{z}^{-1}-\lambda}{1-\lambda\hat{z}^{-1}}$	$\lambda = \frac{\sin\left(\frac{\omega_0-\hat{\omega}_1}{2}\right)}{\sin\left(\frac{\omega_0+\hat{\omega}_1}{2}\right)}$
Highpass $\hat{\omega} > \hat{\omega}_1$	$z^{-1} = -\frac{\hat{z}^{-1}+\lambda}{1+\lambda\hat{z}^{-1}}$	$\lambda = \frac{\cos\left(\frac{\omega_0+\hat{\omega}_1}{2}\right)}{\cos\left(\frac{\omega_0-\hat{\omega}_1}{2}\right)}$
Bandpass $\hat{\omega}_1 < \hat{\omega} < \hat{\omega}_2$	$z^{-1} = -\frac{(\rho-1)-2\lambda\rho\hat{z}^{-1}+(\rho+1)\hat{z}^{-2}}{(\rho+1)-2\lambda\rho\hat{z}^{-1}+(\rho-1)\hat{z}^{-2}}$	$\lambda = \frac{\cos\left(\frac{\hat{\omega}_2+\hat{\omega}_1}{2}\right)}{\cos\left(\frac{\hat{\omega}_2-\hat{\omega}_1}{2}\right)}, \rho = \cot\left(\frac{\hat{\omega}_2-\hat{\omega}_1}{2}\right)\tan\left(\frac{\omega_0}{2}\right)$
Bandstop $\hat{\omega}_1 \not< \hat{\omega} \not< \hat{\omega}_2$	$z^{-1} = \frac{(1-\rho)-2\lambda\hat{z}^{-1}+(\rho+1)\hat{z}^{-2}}{(\rho+1)-2\lambda\hat{z}^{-1}+(1-\rho)\hat{z}^{-2}}$	$\lambda = \frac{\cos\left(\frac{\hat{\omega}_2+\hat{\omega}_1}{2}\right)}{\cos\left(\frac{\hat{\omega}_2-\hat{\omega}_1}{2}\right)}, \rho = \tan\left(\frac{\hat{\omega}_2-\hat{\omega}_1}{2}\right)\tan\left(\frac{\omega_0}{2}\right)$

Noble Identities

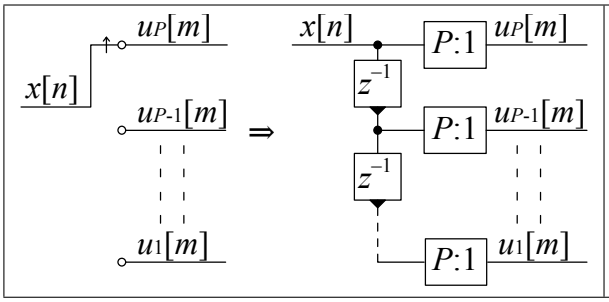
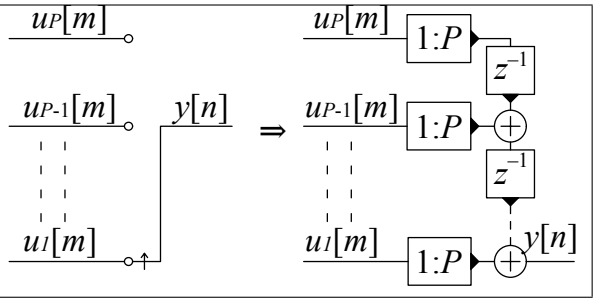


Multirate Spectra

Upsample: $\quad v[n] \xrightarrow{1:Q} x[r] \quad \Rightarrow \quad x[r] = \begin{cases} v\left[\frac{r}{Q}\right] & \text{if } Q \mid r \\ 0 & \text{if } Q \nmid r \end{cases} \quad \Rightarrow \quad X(z) = V(z^Q)$

Downsample: $\quad v[n] \xrightarrow{Q:1} y[m] \quad \Rightarrow \quad y[m] = v[Qm] \quad \Rightarrow \quad Y(z) = \frac{1}{Q}\sum_{k=0}^{Q-1} V\left(e^{\frac{-j2\pi k}{Q}} z^{\frac{1}{Q}}\right)$

Multirate Commutators

Input Commutator	Output Commutator
	

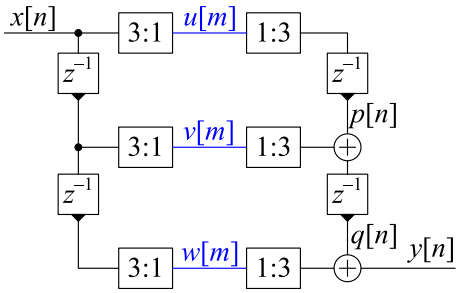
Question 1 [40 Marks]

Q1-1 **Multirate Systems.**

A certain multirate DSP system is described in the block diagram. Given that the output of this system is:

$$y[n] = [\mathtt{a}, \mathtt{b}, \mathtt{c}, \mathtt{d}, \mathtt{e}, \mathtt{f}, \mathtt{g}, \mathtt{h}, \mathtt{i}].$$

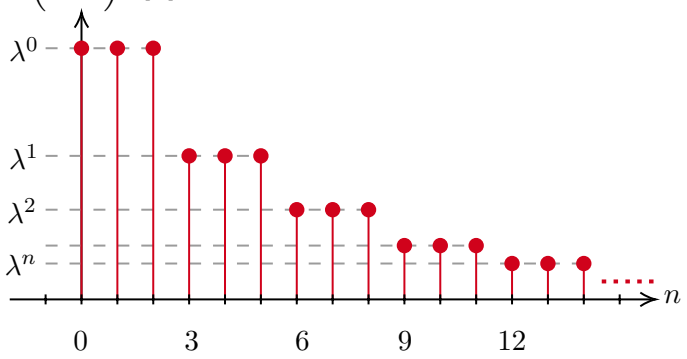
Identify each of the signals:
$x[n], u[m], p[n], v[m], q[n], w[m]$.



Notation: $y[n] = [\mathtt{a}, \mathtt{b}, \mathtt{c}, \mathtt{d}, \mathtt{e}, \mathtt{f}]$ means that $y[0] = \mathtt{a}$, $y[1] = \mathtt{b}$ and so on, and that $y[n] = 0$ outside this range. **[6 Marks]**

Q1-2 The impulse response of a certain filter is shown below.

$$h_\lambda[n] = \left(\lambda^{\lfloor \frac{n}{3} \rfloor}\right) u[n],\ \lambda < 1$$



(a) Compute the z-transform, $H_\lambda(z)$ of $h_\lambda[n] = \left(\lambda^{\lfloor n/3 \rfloor}\right) u[n]$, $\lambda < 1$. **[4 Marks]**

(b) Show that when $\lambda = 0$, the z-transform, $H_0(z)$ reduces to, $\left(1 + z + z^2\right)/z^2$. **[2 Marks]**

(c) Compute the impulse response of $H_0(z)$. **[1 Mark]**

Q1-3 **Continuous to Digital Filter.** Let $H(s)$ denote a second order continuous-time filter:

$$H(s) = \frac{1}{s^2 + \frac{s}{2} + 1}.$$

Obtain the corresponding digital filter $H_d(\omega)$.

Hint: Use the bilinear mapping $s = \frac{(z+1)}{(z-1)}$. You can assume the gain to be unity. **[3 Marks]**

Q1-4 Let $\{x\}$ denote the fractional part of x, that is, $\{2.01\} = 0.1$ and so on.

(a) Plot $f(x)$ for the finite interval $0 \leqslant x \leqslant 3$. **[1 Marks]**

(b) Compute the Fourier transform of the function $f(x) = \{x\}$. **[4 Marks]**

Q1-5 Given z-transform

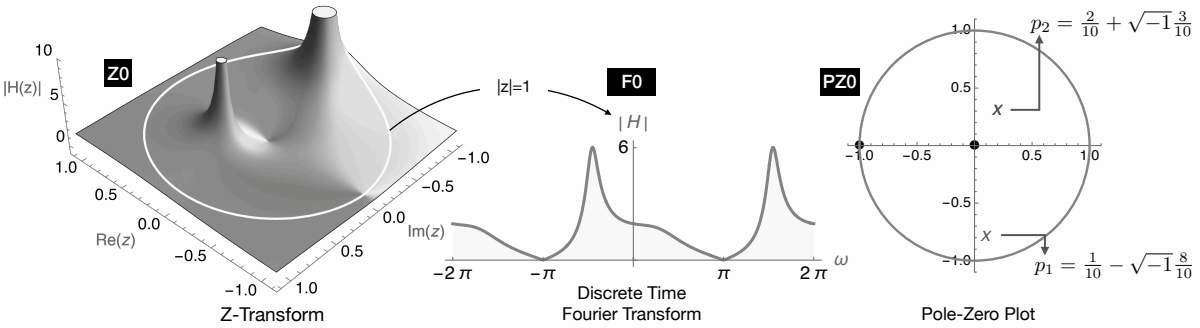
$$H(z) = \frac{2 + 2z^{-1}}{-5 + 0.5z^{-1} + z^{-2}}$$

(a) Draw the zero-pole plot. **[2 Marks]**.

(b) Compute $|H(e^{\jmath\omega})|$ at $\omega = 1.2$. **[2 Marks]**.

(c) Assuming that the system is causal, compute its impulse response. **[3 Marks]**.

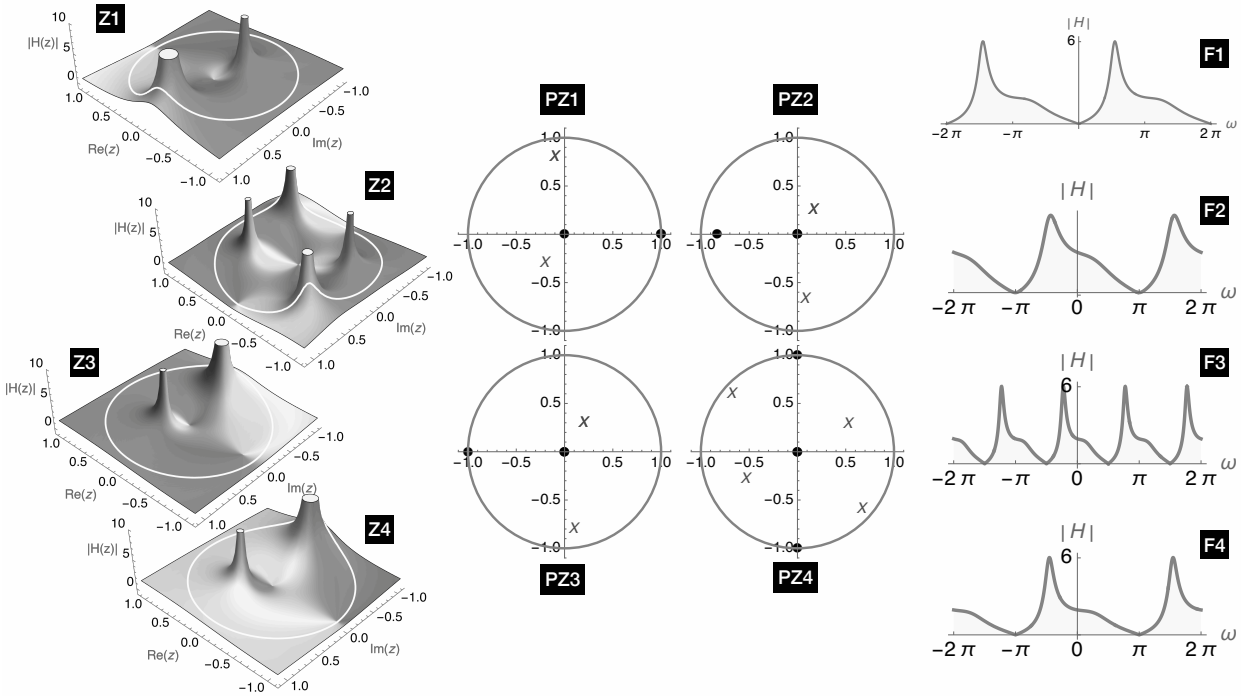
Q1-6 For a certain transfer function, its z-transform, discrete time Fourier transform (DTFT) and pole-zero plot are given by:



In the above, Z0-PZ0-F0 form a matching triplet.

Following this, from the figure below, match the z-transforms (Z1-Z4) and DTFT (F1-F4) with their respective pole-zero plots (PZ1-PZ4). Provide a brief reasoning for your answer.

[8 Marks]



Q1-7 A causal sequence $f[n]$ with *non-minimum phase* property has a z-transform:

$$F(z) = \frac{\prod_{m=1}^{m=4} \left(1 - f_m z^{-1}\right)}{\left(1 - z^{-1}\right)^2 \left(1 - \frac{1}{8} z^{-1}\right)} \quad \text{where} \quad \begin{bmatrix} f_1 & f_2 \\ f_3 & f_4 \end{bmatrix} = \begin{bmatrix} -3/2 & 1/4 \\ 1/3 & -5/2 \end{bmatrix}.$$

Determine the conditions on μ so that the sequence $g[n] = \mu^n f[n]$ is a *minimum phase* sequence.

Hint: For $g[n]$ to satisfy minimum phase property, all of its poles/zeros should be inside the unit circle. **[4 Marks]**

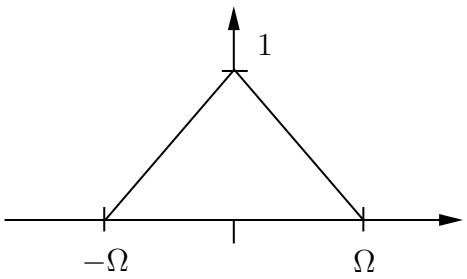


Fig. 1: $X(e^{j\omega}) : \Omega < \pi$.

Question 2 [30 Marks]

Q2-1 The sequence $y[n]$ is the output of the system,

$$x[n] \rightarrow \boxed{\boxed{1:K}} \rightarrow v[n] \rightarrow \boxed{\boxed{Q:1}} \rightarrow y[n]$$

to the input sequence $x[n]$.

(a) Write down the expression for sequences $v[n]$ and $y[n]$ in terms of $x[n]$. **[2 Marks]**

(b) Compute the Discrete-time Fourier transform (DTFT) of $v[n]$ and $y[n]$. **[2 Marks]**

(c) Given that the input $x[n]$ is bandlimited with its DTFT shown in Fig. 1.

i. Let $K = 3$ and $Q = 2$. In order to reconstruct the waveform of the input $x[n]$, we need to low-pass filter the output $y[n]$. Compute the cut-off frequency of this low-pass filter. **[2 Marks]**

ii. Given $K = 3$ and the DTFT of $x[n]$ characterized in Fig. 1, compute the maximum downsampling factor such that there is no spectrum aliasing in y. **[2 Marks]**

Q2-2 A DSP system is depicted in Fig. 2.

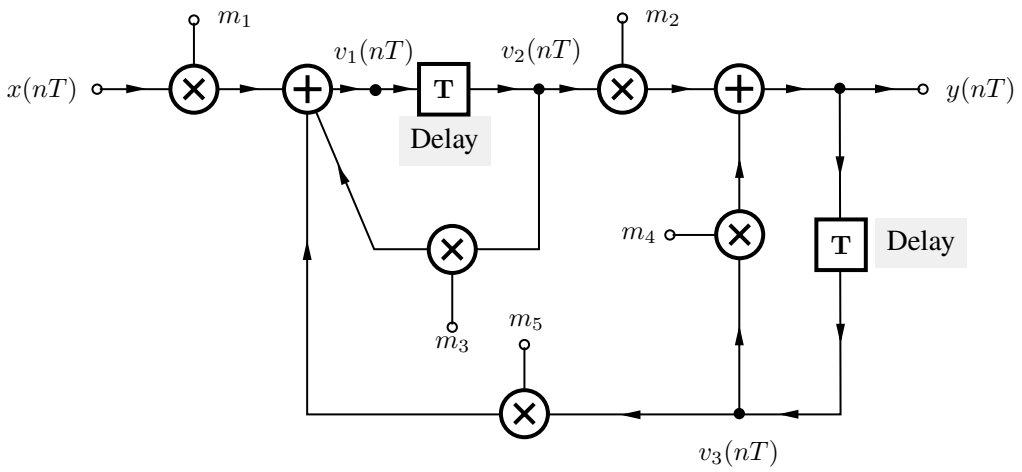


Fig. 2: System Model for Q2-2. T denotes delay by sample step T, $\otimes$ denotes *gain* or *multiplication* and $\oplus$ denotes *addition*.

(a) Write down the time-domain relation between input and output. **[3 Marks]**

(b) Write down the frequency response of this system. **[3 Marks]**

(c) Compute the zeros and poles of the filter system. **[2 Marks]**

(d) Identify the conditions for the system to be stable. **[2 Marks]**

Q2-3 The input-output relation of a system is specified by,

$$\sum_{r=0}^{M} a\left[r\right] y\left[n-r\right] = \sum_{r=0}^{M} b\left[r\right] x\left[n-r\right],$$

where $b\left[n\right] = a^{*}\left[M-n\right]$.

(a) Show that the magnitude response of the transfer function is unity. **[3 Marks]**

(b) Write down z-transform $H(z)$ for 1st-order filter ($M = 1$) with $a[n] \in \mathbb{R}$. **[2 Marks]**

(c) Show that the group delay for the filter,

$$H(z) = \frac{z^{-1} - \alpha^{*}}{1 - \alpha z^{-1}}$$

is always non-negative. **[7 Marks]**

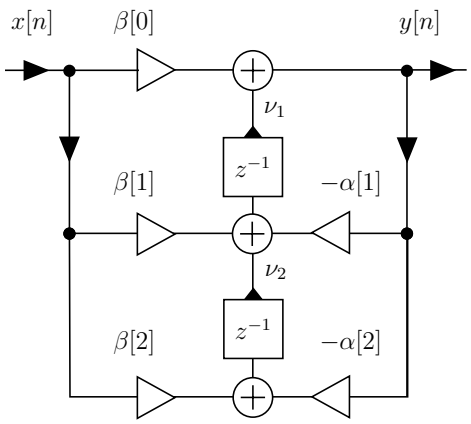


Fig. 3: Filter structure.

Question 3 [30 Marks]

Q3-1 A filtering system is characterized by the block diagram shown in Fig. 3.

(a) Write down the difference equation describing the relationship between $x[n]$ and $y[n]$. **[3 Marks]**

(b) Compute the transfer function $H(z)$ of this filter. **[3 Marks]**

(c) Find the conditions that make the system stable. **[2 Marks]**

Q3-2 Consider the system with the following impulse response

$$h\left[n\right] = \left(\mu\right)^{n+\pi} u\left[n-K\right],$$

where $u\left[n\right]$ denotes the step function and $\mu < 1$.

(a) Derive the phase of the system. **[3 Marks]**

(b) Derive the expression of the group delay $\tau_H\left(e^{\jmath\omega}\right)$. **[4 Marks]**

(c) Find μ, K such that $\tau_H\left(e^{\jmath\pi}\right) = 3$ and $\tau_H\left(e^{\jmath 0}\right) = 1$. **[5 Marks]**

Q3-3 **Multirate System**

Consider the following system with input $x\left[n\right]$ and output $y\left[n\right]$.

$$x\left[n\right] \rightarrow \boxed{1:3} \rightarrow \boxed{2:1} \rightarrow \boxed{\frac{z^{-12}}{\alpha - z^{-12} + \beta z^{-24}}} \rightarrow \boxed{2:1} \rightarrow \boxed{1:3} \rightarrow y\left[n\right]$$

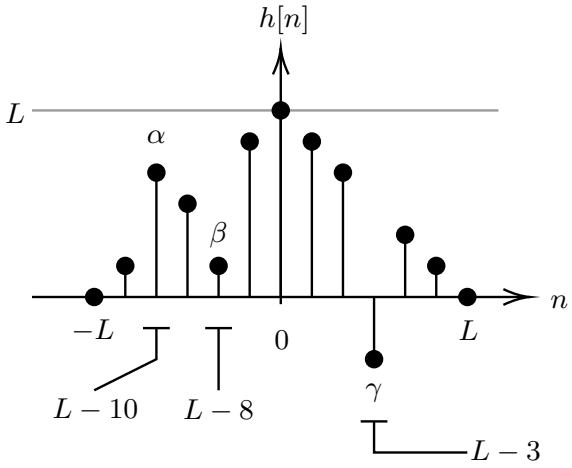
Suppose that an equivalent representation of the above system is given by,

$$x\left[n\right] \rightarrow \boxed{4:1} \rightarrow \boxed{\frac{B\left(z\right)}{A\left(z\right)}} \rightarrow \boxed{1:9} \rightarrow y\left[n\right].$$

Identify α and β if $A\left(z\right)$ has roots at $1/2$ and $1/3$. **[10 Marks]**

Question 4 [30 Marks]

Q4-1 Let $h[n]$ be the filter given by,



while it is given that $L = 6$.

(a) Let $g[n] = d_2 * h[n]$, where $d_2 = d_1 * d_1$, and $d_1[n] = \delta[n] - \delta[n-1]$. Find α, β, γ so that,

$$g[n] = \delta[n].$$

[3 Marks]

(b) Derive the expression of $G(j\omega)$, the DTFT of $g[n]$. **[3 Marks]**

(c) Find the analytical relationship between $G(j\omega)$ and $H(j\omega)$, the DTFT of $h[n]$. **[3 Marks]**

(d) Using the convolution theorem for the DTFT evaluate $H(j\omega)$. **[3 Marks]**

Q4-2 A certain signal processing system operating at sampling rate T, is described by the model in Fig. 4. The system is initially relaxed, that is, $y(nT) = 0$ for $n < 0$, and p is a real constant.

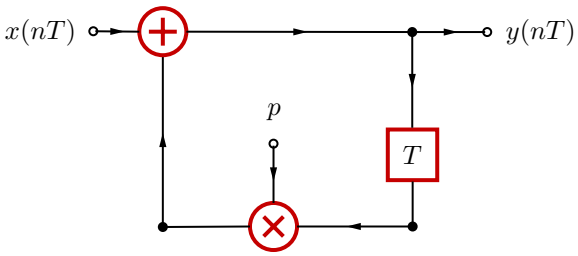


Fig. 4: System Model for Q4-3. T denotes delay by sample step T, $\otimes$ denotes *gain* or *multiplication* and $\oplus$ denotes *addition*.

(a) Find the transfer function? **[2 Marks]**

(b) Find the impulse response of the system. **[2 Marks]**

(c) Find the unit-step response ($x[n] = u[n]$) of the system. **[3 Marks]**

(d) Find the response of the above system to the exponential excitation, $x(nT) = u(nT)e^{j\omega nT}$. **[3 Marks]**

Q4-3 Suppose that a certain signals is defined by ($|\mu| < 1$),

$$h[n] = \left(\sum_{|m| \leqslant n} \mu^{|m|} \right) u[n],$$

where $u[n]$ is the usual unit-step sequence. Show that its Fourier transform is given by,

$$H(e^{\jmath\omega}) = \frac{1 + \mu e^{-\jmath\omega}}{(1 - e^{-\jmath\omega})(1 - \mu e^{-\jmath\omega})}.$$

[8 Marks]